

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1104

COURSE OUTCOME COVERED

***CO1:** Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

Assessment Tool Weightage: 60%

Dr Ramamoorthy

QUESTIONS: 12

Total Marks:60

Annexure A

CONCEPT MAPPING

Code of Conduct:

I N. Hima Harshitha (192371063) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N. Hima Harshitha

CONCEPT MAPPING

1. Construct a Concept Map

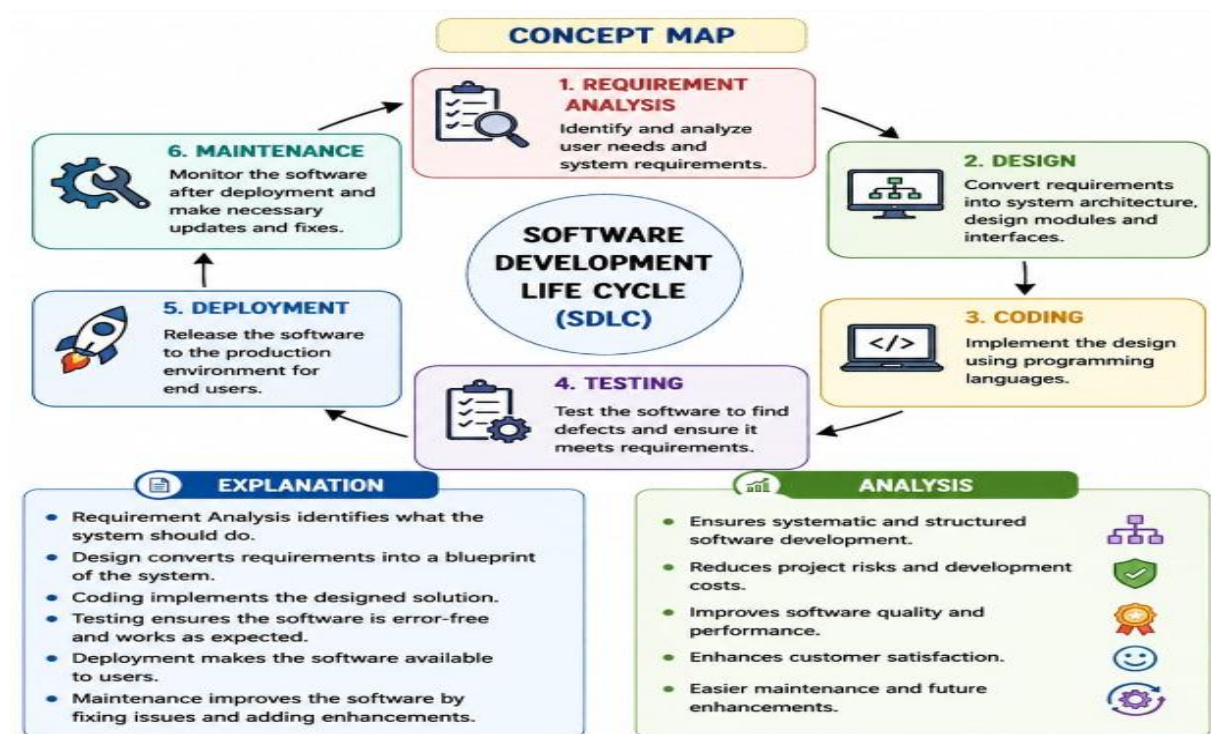
Given:

Requirement Analysis → _____ → Coding → Testing → _____ → Maintenance

Answer:

The missing concepts are:

- Design
- Deployment



Explanation:

- Design converts software requirements into a detailed system architecture and design plan.
- Deployment releases the tested software to end users for use in the real environment.

Analysis:

- Ensures a systematic and organized software development process.
- Reduces project risks by following well-defined stages.
- Improves software quality, reliability, and long-term maintainability.

2. Construct a Concept Map

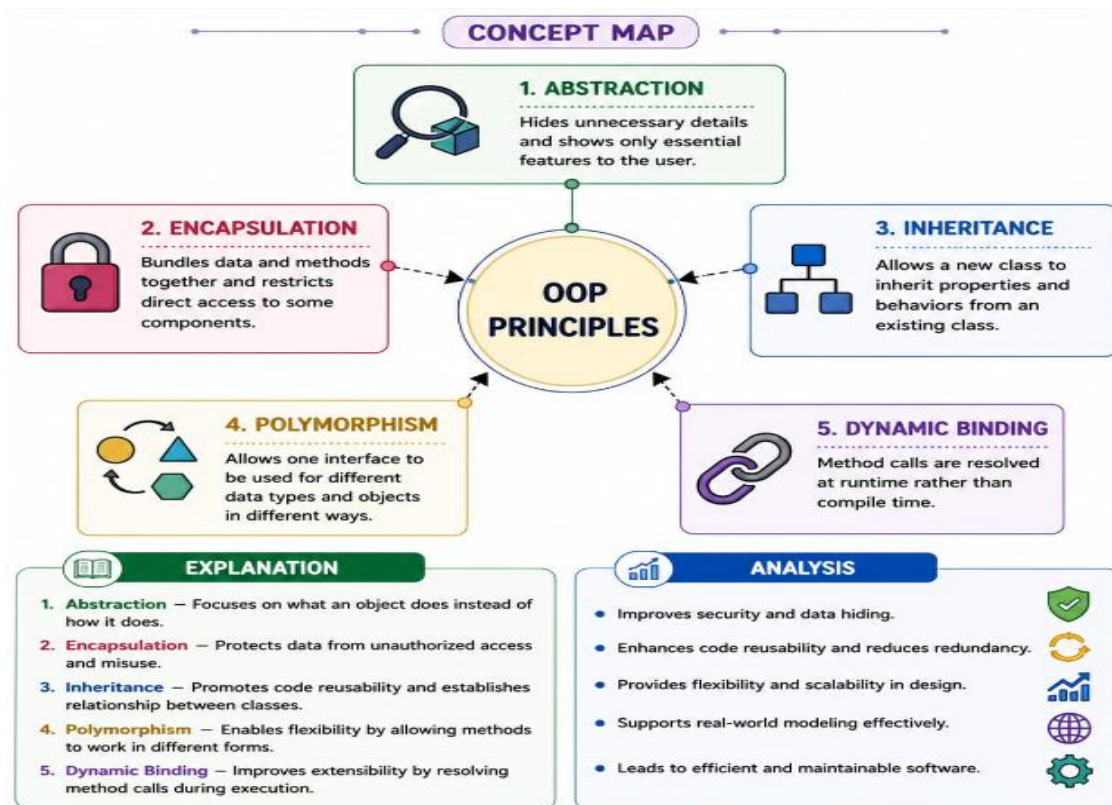
Given:

Object-Oriented Principles → Abstraction → _____ → Inheritance → _____

Answer:

The missing concepts are:

- Encapsulation
- Polymorphism



Explanation:

- Encapsulation protects data from unauthorized access.
- Polymorphism allows one interface to support multiple behaviours.

Analysis:

- Improves security and flexibility.
- Enhances code reusability.
- Supports efficient software design.

3. Construct a Concept Map

Given:

Class → Blueprint

Object → Instance

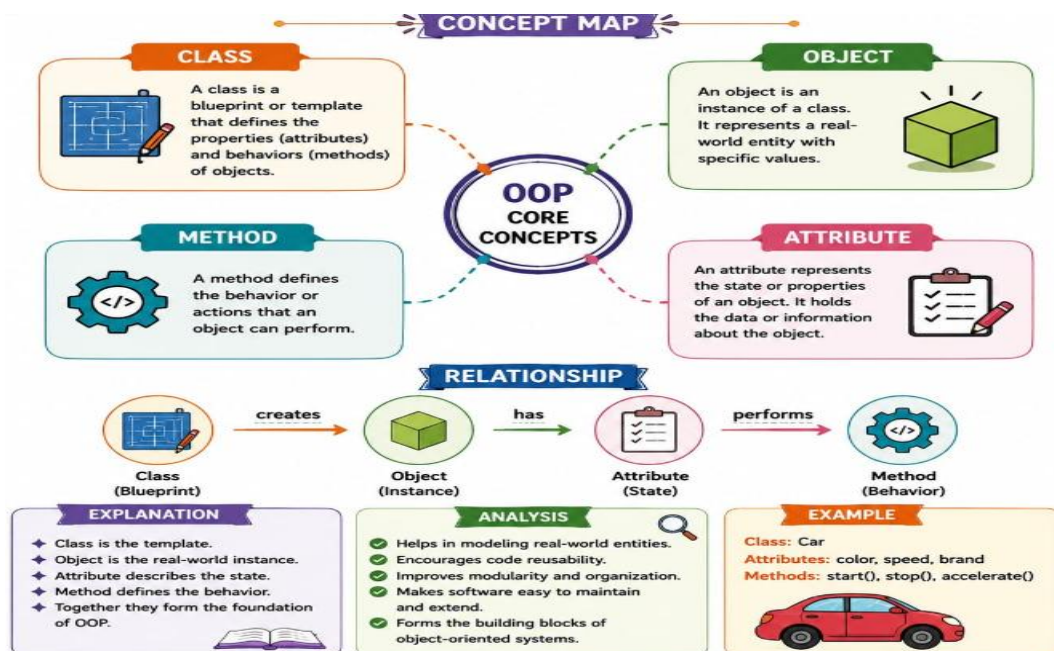
Method → _____

Attribute → _____

Answer:

The missing concepts are:

- Behavior
- State



Explanation:

- Methods define the behaviour of objects.
- Attributes represent the state of objects.

Analysis:

- Helps model real-world entities.
- Defines object characteristics and actions.
- Improves system organization.

4. A Concept Map Shows

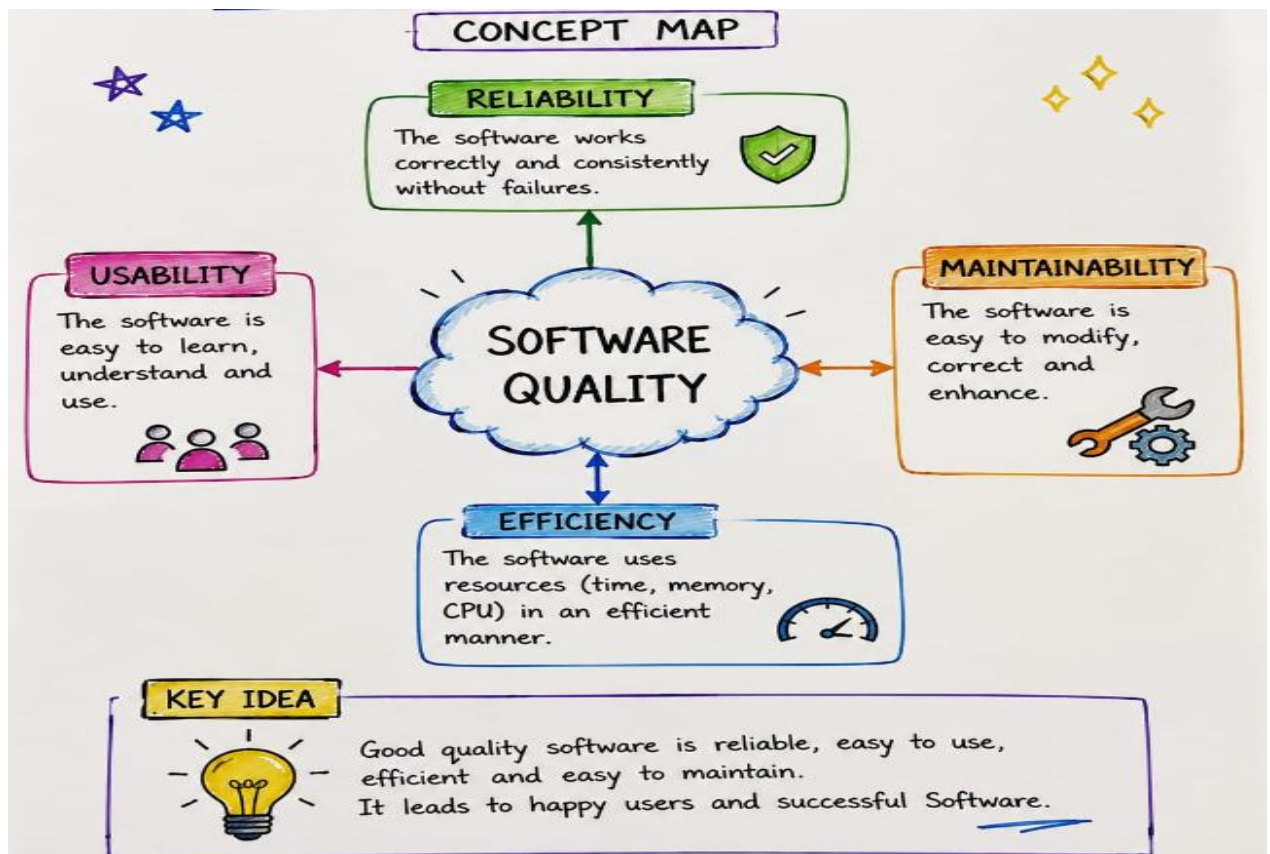
Given:

Software Quality → Reliability → _____ → Maintainability

Answer:

The missing concept is:

- Usability



Explanation:

- Usability measures how easily users interact with software.
- Reliable software performs consistently.
- Maintainable software is easy to update.

Analysis:

- Improves user satisfaction.
- Enhances product acceptance.
- Contributes to software success.

5. UML Concept Map

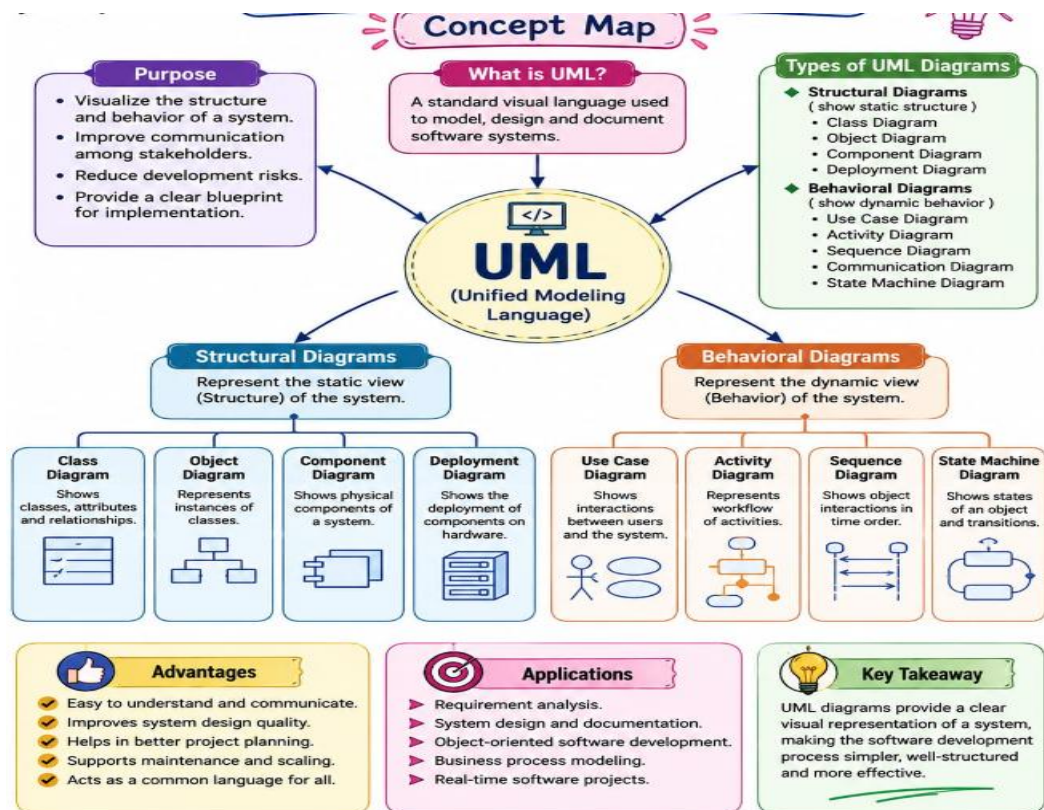
Given:

UML → Structural Diagrams → _____ → Behavioral Diagrams → Activity Diagram
→ Interaction Diagrams → _____

Answer:

The missing concepts are:

- Object Diagram
- Communication Diagram



Explanation:

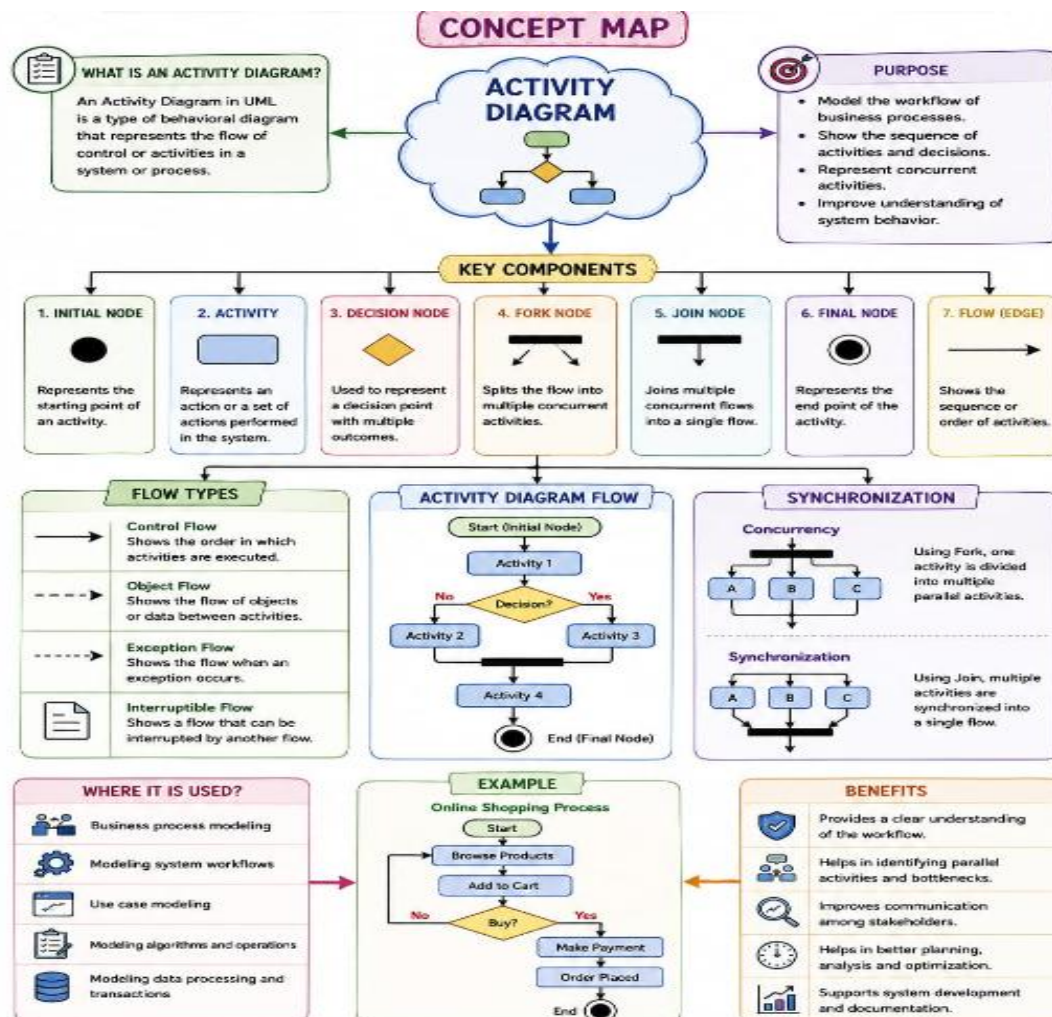
- Object diagrams represent object instances and relationships.
- Communication diagrams show message exchange among objects.

Analysis:

- Improves system visualization.
- Supports object-oriented analysis.
- Facilitates stakeholder communication.

6. Given:

Requirements ✓ → Design ✓ → Coding ✗ → Testing ✗ → Deployment ✗



Explanation:

- Failure occurs during the Coding phase.
- Program logic contains implementation errors.

Analysis:

- Errors affect subsequent phases.
- Testing cannot proceed effectively.
- Software quality decreases.

Solution:

- Review source code.
- Apply coding standards.
- Conduct code inspections before testing.

7. Construct a Concept Map

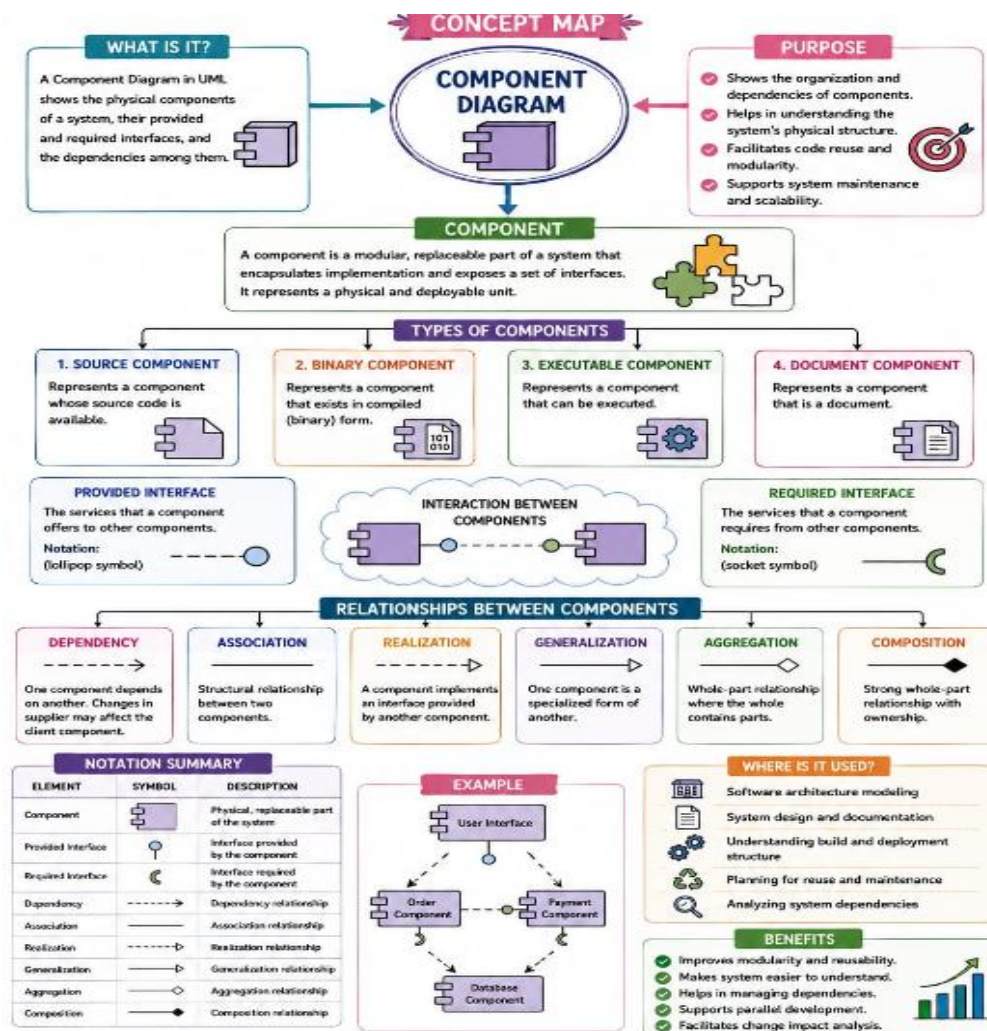
Given:

Problem Identification → _____ → Design → Implementation → _____

Answer:

The missing concepts are:

- Requirement Analysis
- Testing



Explanation:

- Requirement Analysis identifies user needs and system requirements.
- Testing verifies whether the implemented system meets the specified requirements.

Analysis:

- Improves software quality.
- Reduces development errors.
- Ensures successful project completion.

8. Construct a Concept Map

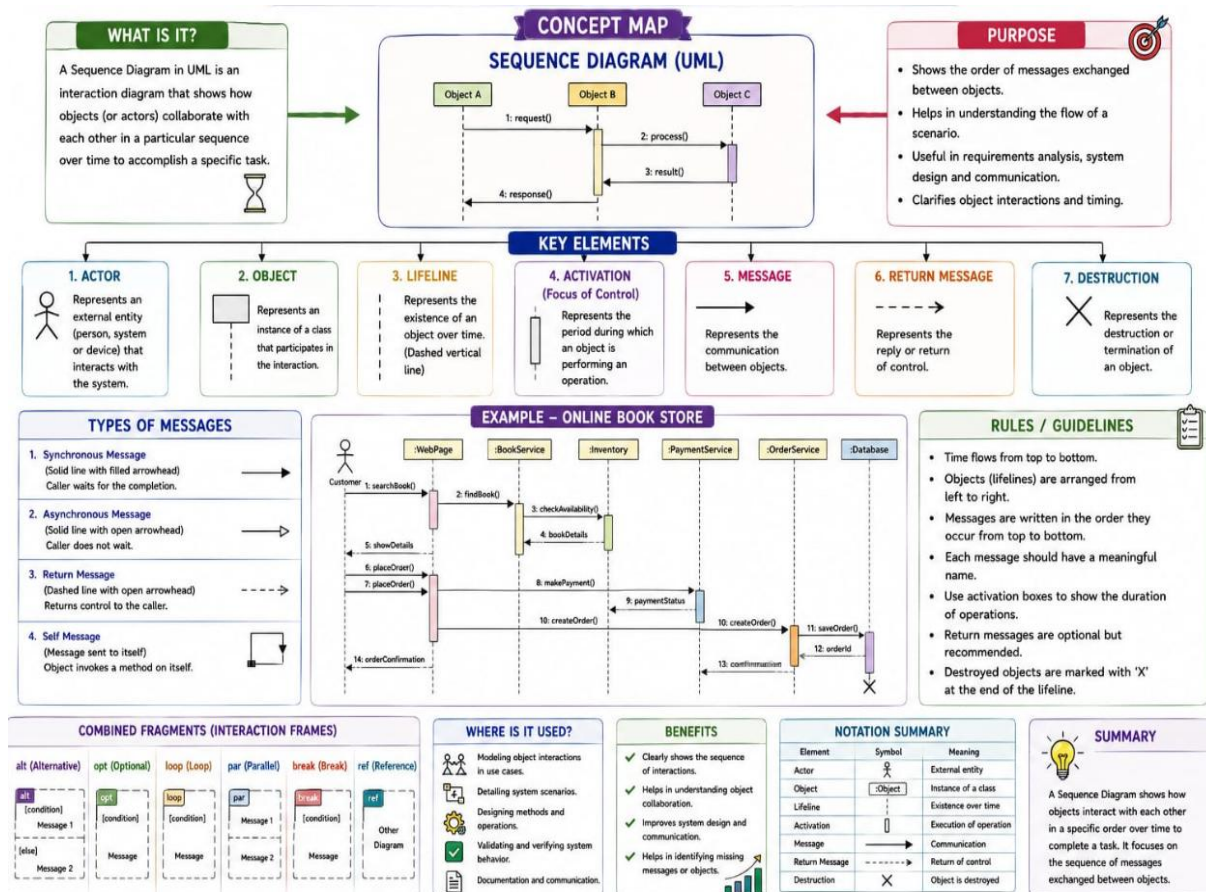
Given:

Object-Oriented Features → Class → _____ → Message Passing → _____

Answer:

The missing concepts are:

- Object
- Dynamic Binding



Explanation:

- Objects are instances of classes.
- Dynamic binding allows method calls to be resolved at runtime.

Analysis:

- Supports flexibility in software design.
- Improves code maintainability.
- Enables runtime polymorphism.

9. Construct a Concept Map

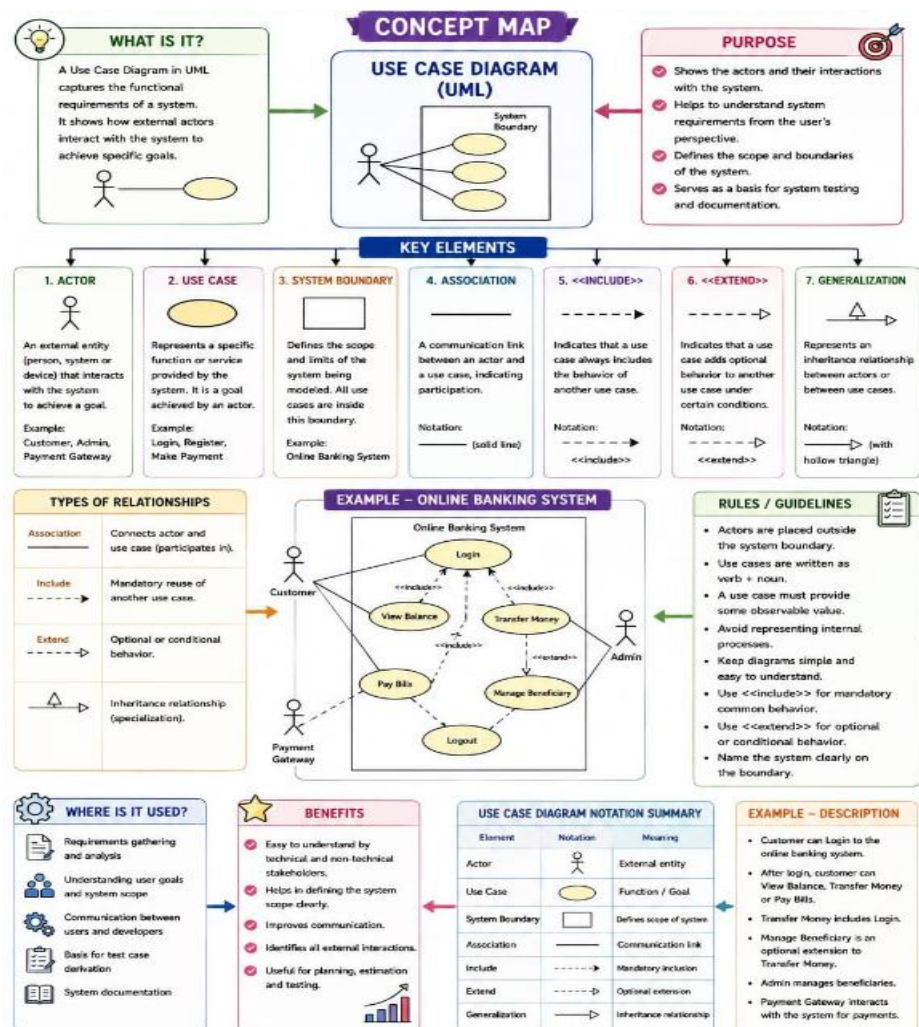
Given:

Software Development Models → Waterfall → _____ → Spiral → _____

Answer:

The missing concepts are:

- Incremental Model
- Agile Model



Explanation:

- Incremental development delivers software in stages.
- Agile supports iterative development with continuous customer feedback.

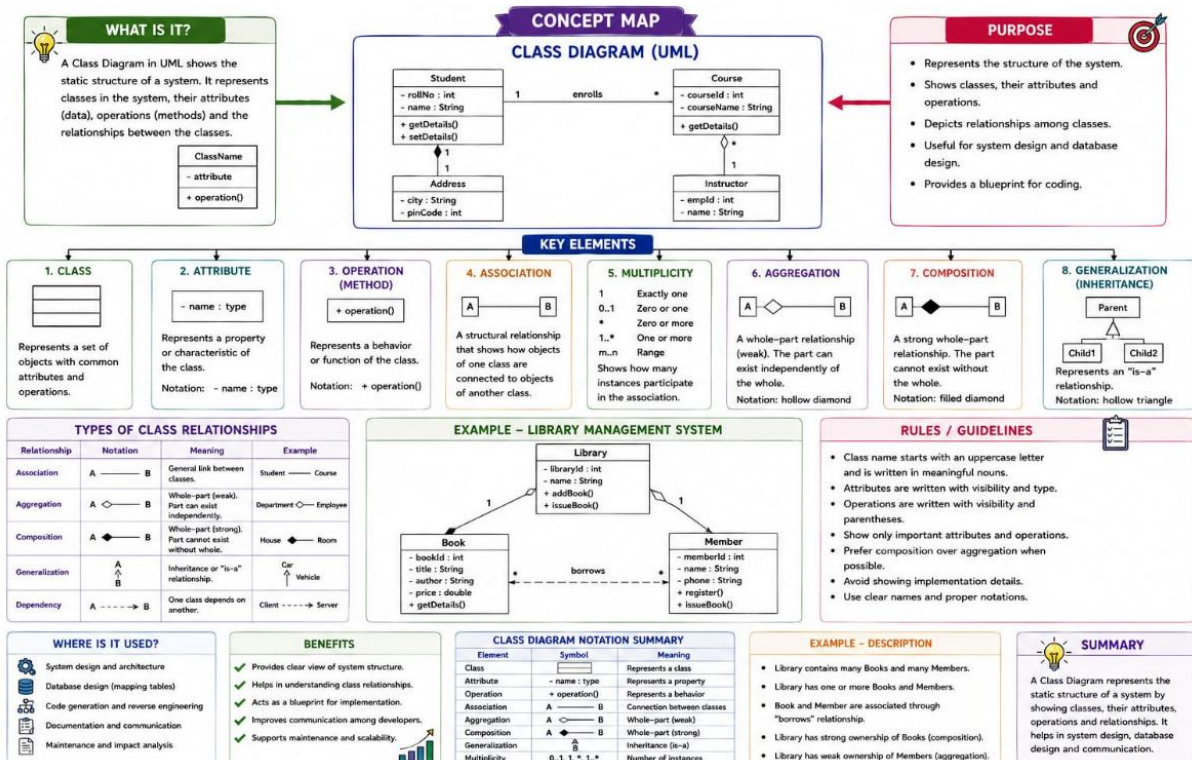
Analysis:

- Enhances adaptability.
- Reduces project risks.
- Improves customer satisfaction.

10. Construct a Concept Map

Given:

Class Diagram → Class → _____ → Association → _____



Answer:

The missing concepts are:

- Attribute
- Method

Explanation:

- Attributes define the properties of a class.
- Methods define the operations performed by a class.

Analysis:

- Represents object structure effectively.
- Improves software documentation.
- Supports object-oriented design.

11. Construct a Concept Map

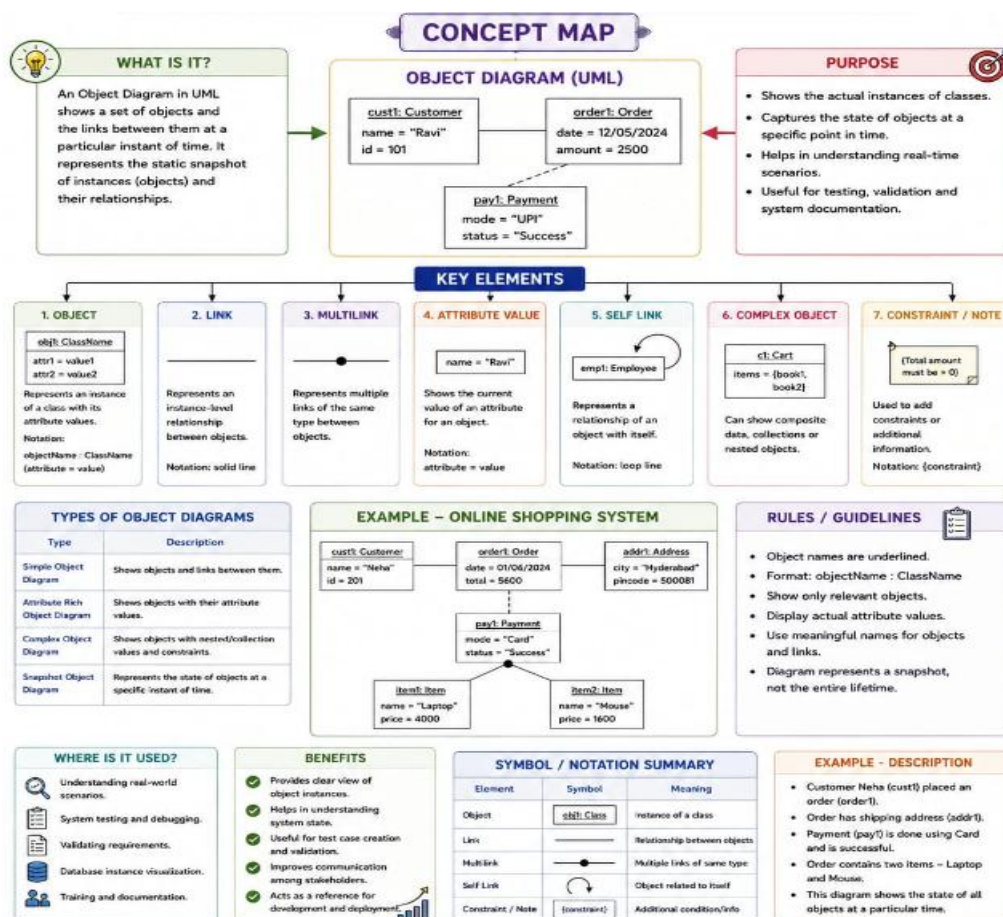
Given:

Software Engineering Principles → Modularity → _____ → Reusability → _____

Answer:

The missing concepts are:

- Maintainability
- Scalability



Explanation:

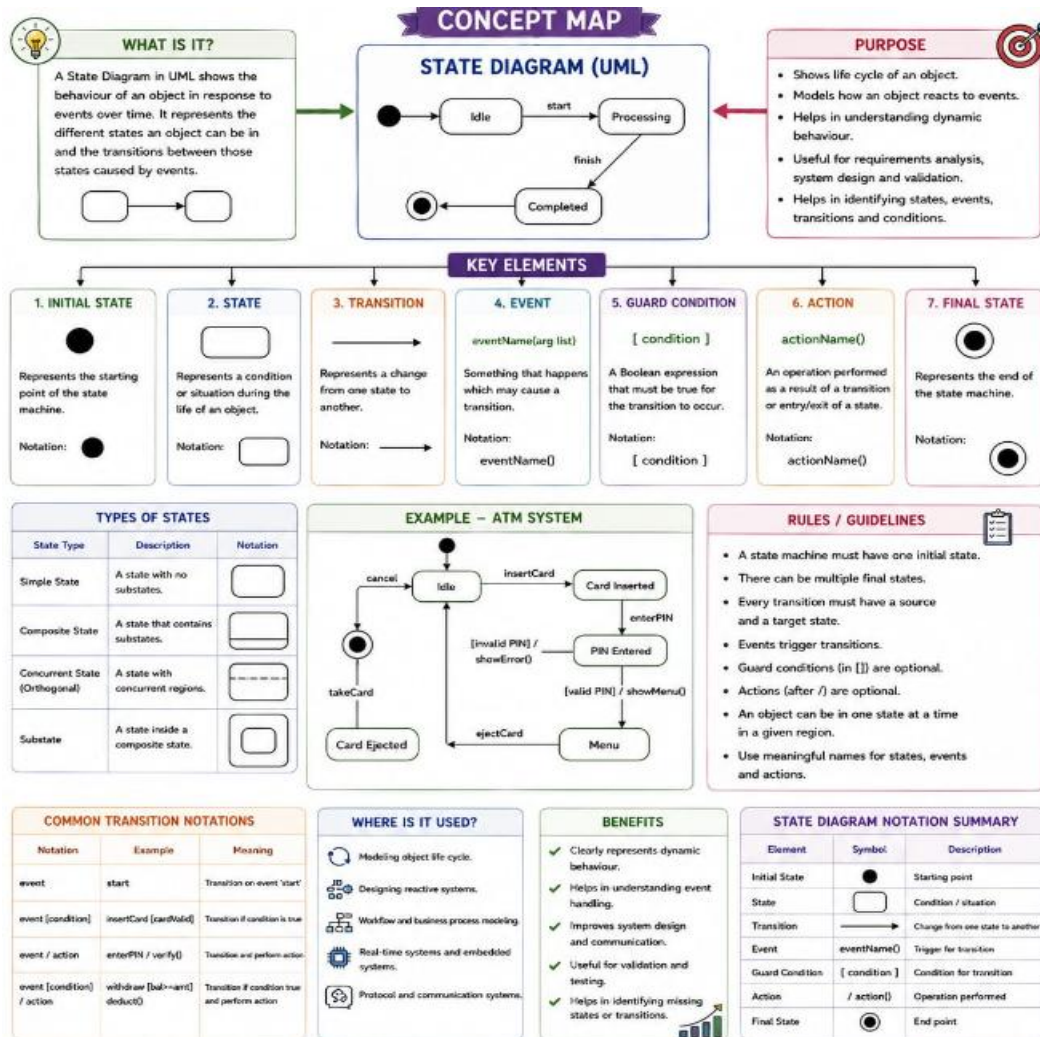
- Maintainability simplifies software updates.
- Scalability enables systems to handle increased workload efficiently.

Analysis:

- Improves software performance.
- Reduces maintenance effort.
- Supports long-term software evolution.

12. Given:

Requirement Analysis ✓ → System Design ✓ → Coding ✓ → Integration ✗ → Deployment ✗



Explanation:

- Failure occurs during the Integration phase.
- Individual modules fail to communicate correctly.

Analysis:

- Causes functional inconsistencies.
- Delays software deployment.
- Increases debugging effort.

Solution:

- Perform integration testing.
- Verify module interfaces.
- Resolve compatibility issues before deployment.

